A CECH-LOCAL PROPERTY HOLDS GLOBALLY ONCE IT HOLDS ON AFFINE FLAT FINITELY PRESENTED OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Surjective etale presentation. existence of a disjoint union of affines U -> X surjective etale (cross-chapter, structural). The covering through which Cech descent is run.
- Separatedness of etale morphisms. etale morphisms of algebraic spaces are separated (cross-chapter, structural). Makes all Cech fibre products separated, hence their affine etale covers have affine fibre products.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let P be a property of objects in $(Spaces/X)_{fppf}$ such that whenever $\{U_i \to U\}$ is a covering in $(Spaces/X)_{fppf}$, then*

$$P(U_{i_0} \times_U \ldots \times_U U_{i_p}) \text{ for all } p \geq 0,\ i_0, \ldots, i_p \in I \Rightarrow P(U)$$

If $P(U)$ for all U affine and flat, locally of finite presentation over X, then $P(X)$.

Proof. Let U be a separated algebraic space locally of finite presentation over X. Then we can choose an étale covering $\{U_i \to U\}_{i \in I}$ with V_i affine. Since U is separated, we conclude that $U_{i_0} \times_U \ldots \times_U U_{i_p}$ is always affine. Whence $P(U_{i_0} \times_U \ldots \times_U U_{i_p})$ always. Hence $P(U)$ holds. Choose a scheme U which is a disjoint union of affines and a surjective étale morphism $U \to X$. Then $U \times_X \ldots \times_X U$ (with $p+1$ factors) is a separated algebraic space étale over X. Hence $P(U \times_X \ldots \times_X U)$ by the above. We conclude that $P(X)$ is true. □

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